

# Modeling the Dynamics of Collective Intelligence Under Stress: An AI-Based Framework for Predicting Adaptation and Breakdown in Human–AI Groups

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**Abstract**—Collective intelligence is usually measured as a single score, taken once, under calm conditions. This study treats it instead as a trajectory and asks whether the early interaction record of a group under stress indicates how that group will end up. An agent-based model of a hidden-profile task was built in which agents must pool distributed information items through trust-weighted exchange, stress is a single load parameter that suppresses both sending and accepting, and one human member is removed partway through each run. Five conditions cross stress with the presence of a load-immune, imperfectly reliable AI participant, with a size control. Each of 5,000 runs was labelled stable, adaptive, stressed or broken down from the information state alone, and described by twelve features taken only from the message log. A random forest reached a held-out macro-F1 of 0.34, against 0.21 for a model that knows only the condition, but within any single condition the features did little better than chance. Every broken-down run traced to a required item that only the removed member had held. In this model, whether a group breaks down is settled mainly by which member it loses, not by how it communicated beforehand. The model is a prototype: it fixes the labels, features and effect sizes that a study with human volunteers would need, and it locates the one design choice, random removal of a member, that such a study should replace.

**Index Terms**—collective intelligence, agent-based simulation, human–AI teams, stress, early warning signals, hidden profile

## I. INTRODUCTION

Collective intelligence is the capacity of a group to solve problems more effectively than its members acting alone, and the frameworks proposed for studying it are numerous and only loosely connected [24]. It is a group-level property rather than an aggregate of individual ability: meta-analytic evidence indicates that the quality of a group’s collaboration process predicts collective performance roughly twice as strongly as the skill of its individual members [1]. How a group interacts matters more than who is in it.

Almost all of this evidence, however, comes from quiet rooms. Participants are strangers, sessions last about an hour, and nothing goes wrong. The settings in which collective intelligence actually matters are the opposite: epidemic re-

sponse, flood evacuation, emergency resource allocation. In these environments, groups face uncertainty, information overload, resource scarcity and disrupted communication channels. Under such conditions a group may reorganise and cope, or it may fragment. No current model predicts which.

The question has become more pressing because groups increasingly contain artificial agents. That addition is not obviously beneficial. The largest synthesis available reports that combining humans and AI often produces worse decisions than the better party would have produced alone [2]. Whether this penalty grows or shrinks under stress is unknown, because the underlying experiments were not conducted under stress.

## A. Research Objectives

1) *Simulate collective behaviour under stress*: construct an agent-based model of a group solving a task that requires pooling distributed information, across four conditions crossing stress with AI participation.

2) *Define breakdown operationally*: produce a reproducible definition of collective breakdown grounded in information integration and recovery, independent of the interaction features used for prediction.

3) *Identify structural indicators*: isolate features of the interaction record that precede adaptation or breakdown.

4) *Predict transitions*: train and evaluate models that classify group state from early interaction structure, and report honestly whether adaptation and breakdown are separable at all.

## B. Significance and Innovation

The contribution is to treat collective intelligence as a trajectory rather than a score, and to treat the definition of failure as a research problem rather than an assumption. Existing measurement yields one number describing how a group performed. This study models the group as an evolving interaction structure and asks what state it will occupy next, so that intervention becomes possible before failure rather than diagnosis after it.

### C. Summary of Findings

The model was built, calibrated on reserved seeds, and used to generate 5,000 labelled runs. Stress had a clear effect on outcomes: it almost eliminated early completion and pushed a quarter to a third of runs into prolonged partial integration. Classifiers trained on the early message log beat every naive baseline, but most of what they learned was which condition a run belonged to. Within a condition, and between adaptation and breakdown in particular, they did little better than chance. The reason is structural and is traced in Section V: breakdown in this model is decided at the moment of the perturbation by whether the lost member held something no one else did. Section II reviews the literature, Section III describes the methodology as implemented, Section IV reports results, and Sections V and VI discuss them and set out what should change next. Throughout, the model is treated as a prototype of an experiment with human volunteers, not as a substitute for one; Section VI makes the case for that experiment.

## II. LITERATURE REVIEW

Research bearing on this study is distributed across communities that rarely cite one another: the psychometric measurement of collective intelligence, organisational research on teams under stress, human-AI interaction, network science, and the complex-systems literature on critical transitions. This review synthesises them thematically and locates the gap at their intersection.

### A. Scope and Search Strategy

Searches were conducted in August 2026 across Google Scholar, arXiv, Semantic Scholar and PubMed, with direct retrieval from *Nature*, *Science*, Springer and IEEE Xplore. Representative Boolean strings were "collective intelligence" AND (stress OR crisis OR breakdown), ("human-AI" OR "human-agent") AND (team OR collective) AND performance, ("threat rigidity" OR "team resilience") AND coordination, and ("agent-based" OR simulation) AND (validation OR "early warning"). Backward and forward citation chaining was applied to the foundational works [1], [3].

Peer-reviewed empirical studies, meta-analyses, systematic reviews and methodological papers were included, together with recent preprints where no peer-reviewed equivalent exists. Studies of individual performance without a group comparison, commercial sources and non-peer-reviewed commentary were excluded. Screening proceeded from title to abstract to full text.

### B. Collective Intelligence as a Static Measurement

Woolley et al. [3] administered batteries of tasks to 699 participants working in groups of two to five and identified a single factor,  $c$ , explaining much of the variance in performance across otherwise unrelated tasks. The correlates proved more surprising than the factor itself. Average member intelligence predicted  $c$  only weakly, while the average social sensitivity of

members, the evenness of speaking-turn distribution and the proportion of women in the group predicted it considerably better.

Riedl et al. [1] revisited the question meta-analytically across 22 studies comprising 1,356 groups and 5,279 participants. Evidence for a general collective intelligence factor survived leave-one-out testing, and collaboration-process measures carried approximately twice the predictive weight of individual member skill.

The finding is contested rather than settled. Bates and Gupta [4] reported across three studies with 312 participants that individual IQ accounted for approximately 80 per cent of the variance in group-level performance, with no detectable relationship to gender balance or turn-taking. This study takes no position on that dispute, since its subject is how collective performance *changes* under load rather than whether a general factor exists.

What both camps share is more consequential here. Collective intelligence is treated throughout as a quantity a group possesses: a score obtained from a single session, under conditions deliberately kept calm, among strangers with no prior history. A score describes an endpoint. It cannot describe a trajectory, and it cannot indicate when a group began to fail.

### C. Stress and the Loss of Shared Knowledge

Ellis [5] supplies the mechanism on which this study rests. Across 97 teams performing a command-and-control simulation, acute stress degraded shared mental models and transactive memory—the distributed awareness of who knows what—and that degradation statistically explained the resulting performance loss. The failure was structural rather than motivational: members continued working, but the group's shared representation of its own knowledge came apart. This identifies what to measure. If stress destroys the routing of knowledge rather than the effort of individuals, breakdown should be observable as a failure of information to reach the members who need it.

The wider stress literature is considerably less consistent. Threat-rigidity theory [6] predicts two specific responses to perceived threat: a restriction in information processing and a constriction of control as authority centralises upward. Yet Driskell and Salas [7] tested centralisation against an increased-receptivity hypothesis and found the latter, with participants of both high and low status becoming more willing to defer to a partner's judgement under stress.

Mojzisch et al. [8], reviewing fifty years of experimental work on group decision making under acute stress, attribute much of this inconsistency to methodology. Studies rarely employ validated stress inductions, routinely conflate stress with time pressure, and frequently measure individual choices made inside a group rather than genuine collective decisions. They report no empirical support for groupthink's central claim regarding cohesion, and describe group-level evidence for threat-rigidity as mixed at best. Their recommendations shape this study in two ways. Stress is operationalised as a single controlled parameter rather than conflated with time pressure,

mirroring the separation that validated human instruments such as the group version of the Trier Social Stress Test [9] achieve in laboratory settings. The task possesses an objectively correct answer, following the hidden-profile paradigm of Stasser and Titus [10]. In a hidden profile, the information required to identify the correct option is distributed such that no member can reach it alone, making information sharing both necessary for success and directly measurable.

A separate literature offers a vocabulary for the group's response to that load rather than for the load itself. Stoverink et al. [11] draw on conservation-of-resources theory to model team resilience as a capacity that resides in the team as a whole: built up jointly from resources members hold in common, and unfolding through the team's ongoing processes rather than fixed in advance. Their starting observation is that teams under sustained adversity either recover their functioning or instead see their coordination break down and their output decline, and the model is an attempt to explain what separates the two. That same pairing, adaptation against breakdown, is what the present study makes measurable through the interaction record rather than through resource-based indicators, treating it as a trajectory a group is on rather than a trait it either has or lacks.

#### **D. Human–AI Collectives Under Load**

Peeters et al. [12] argue that in mixed human–AI systems the significant risks emerge at the collective level rather than in any single component, which is also the premise of this study. The empirical record counsels caution. Vaccaro et al. [2] conducted a preregistered systematic review and meta-analysis of 106 experimental studies reporting 370 effect sizes, finding that human–AI combinations performed on average worse than whichever of the two performed better alone. Losses concentrated in decision-making tasks while gains appeared in content creation, and design features in which the field has invested heavily—displaying explanations, showing model confidence—did not moderate the outcome. Two properties of that corpus define the gap this study enters: it is overwhelmingly dyadic, pairing a single person with a single model, and it contains almost no sustained stress.

Burton et al. [13] identify a failure mode specific to the collective level. When many members of a group consult the same small number of language models, privately held beliefs converge and the diversity on which collective intelligence depends erodes before deliberation begins. Premature consensus is particularly dangerous because, observed from outside, it is difficult to distinguish from efficient coordination.

Shiiku et al. [14] demonstrate that these effects are time-dependent. Running an iterative storytelling task across  $5 \times 5$  grid networks under human-only, AI-only and mixed configurations, they found that AI-only networks initially produced work rated more creative and more diverse, but that mixed networks overtook them on diversity across successive iterations. A single measurement taken at either end of that experiment would have supported the opposite conclusion, which constitutes a direct argument for temporal modelling.

Taken together, these findings suggest a mechanism worth testing rather than assuming. An artificial participant does not experience stress and does not reduce its output as load rises. Its contributions therefore occupy a growing share of group communication precisely when human members are least able to scrutinise them. If that agent is imperfectly reliable, stress should amplify the propagation of its errors.

#### **E. Network Structure and Position**

Structure carries substantial explanatory weight independent of composition. Shirado and Christakis [15] showed that autonomous agents behaving with a small degree of randomness improved a human group's ability to solve a distributed coordination problem, but that the magnitude of the benefit depended on where in the network those agents were placed. Centola [16] synthesises the wisdom-of-crowds and collective-problem-solving traditions and concludes that on difficult tasks, decentralised and informationally inefficient networks tend to outperform densely connected ones, because slower information flow protects unusual ideas long enough for them to be evaluated rather than suppressed. Cui and Yasseri [17] formalise such collectives as multilayer networks with distinct cognition, physical and information layers, the closest existing formalism to the representation required here. The size of the units in which people interact also matters: Iacopini et al. [25] show that interactions in groups, rather than in pairs, change how large a committed minority must be before a population tips to a new convention.

These questions are operational rather than abstract in the settings that motivate this work, where AI already supports disease surveillance and emergency logistics [18] and climate action increasingly depends on coordinating distributed citizen, non-governmental and government actors [19].

#### **F. Detecting Transitions Before They Complete**

Two technical literatures supply the modelling apparatus. Temporal graph learning has matured rapidly, with memory-based architectures such as Temporal Graph Networks [20] maintaining per-node state updated by each interaction. These models are well benchmarked, but almost entirely on dynamic link prediction and anomaly detection. Classifying the state of an entire graph as it evolves—which is what the question “is this group stable, adapting, or breaking down?” amounts to—remains comparatively undeveloped.

Early-warning-signal theory supplies the other half. George et al. [21] review critical slowing down: as a system approaches a bifurcation it recovers more slowly from perturbation, and this appears in observable time series as rising variance and rising lag-1 autocorrelation. The phenomenon has been demonstrated across ecosystems, financial markets, climate records and the onset of epileptic seizures. Rietkerk et al. [22] add a qualification that any applied system must handle: the same signals also precede transitions that are not catastrophic, so their presence indicates an approaching change of regime rather than an approaching failure. Distinguishing the two is part of the problem rather than a detail.

TABLE I  
RESEARCH GAPS AND CORRESPONDING DESIGN DECISIONS

| Gap identified                                                   | How this study responds                                                        |
|------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Measured as a static score under calm conditions [1], [3]        | Interaction structure observed across time under rising stress                 |
| Stress damage inferred from outcomes, not observed [5]           | Integration measured directly: which required items reach a decision-maker     |
| Human–AI evidence dyadic, unstressed [2], [13]                   | Multi-agent groups with a load-immune, imperfectly reliable AI participant     |
| Structure known to matter but not tracked under load [15], [16]  | Full message log retained; structure computed from it over any window          |
| No operational definition of breakdown, hence no training labels | Breakdown defined from integration after a controlled perturbation             |
| Simulated data lacks validation [23]                             | Thresholds fixed on reserved seeds; labels and features kept strictly separate |

The relevance here is methodological as well as theoretical. If recovery time distinguishes systems near collapse from systems merely disturbed, then adaptation and breakdown—which resemble one another closely at the moment of disruption, since both involve structural reorganisation—may be separable by what follows the disruption rather than by the disruption itself. This motivates introducing a controlled perturbation mid-run and measuring whether information flow resumes.

#### G. Validity Constraints on Simulated Data

Because the data for this study are generated by simulation, one methodological constraint is binding. Larooij and Törnberg [23] reviewed generative agent-based models and concluded that validation, not expressiveness, is the field’s central problem. They document black-box model structure, cultural bias inherited from training data, unpredictable behaviour on out-of-distribution scenarios, and validation procedures in which a model effectively adjudicates the plausibility of its own output. Their argument is not that such simulations are useless, but that a simulated dataset carries no evidential weight unless its validation strategy is fixed in advance.

This study accepts that standard. Labelling criteria and thresholds were fixed on a reserved set of calibration seeds before the study dataset was generated, and were not revised after any classifier was trained; sensitivity across threshold values is reported; and the agent rules are deliberately simple and inspectable rather than language-model-based, so that any observed transition can be traced to the rule that produced it.

#### H. Research Gaps and Unique Contributions

Table I maps the gaps identified above onto the design decisions that follow from them. Four contributions result.

1) *Structural observation under load*: the group is represented as a temporal interaction structure and observed while stress is applied, so that degradation is measured in the coordination itself rather than inferred from a final score.

2) *An operational definition of breakdown*: a group is classified as broken down when the information required

TABLE II  
EXPERIMENTAL CONDITIONS

| Condition                 | Humans | AI | Stress          |
|---------------------------|--------|----|-----------------|
| C1 Human-only, calm       | 12     | 0  | 0.10 throughout |
| C2 Human-only, stressed   | 12     | 0  | 0.10 → 0.32     |
| C3 Human+AI, calm         | 12     | 1  | 0.10 throughout |
| C4 Human+AI, stressed     | 12     | 1  | 0.10 → 0.32     |
| C5 Size control, stressed | 13     | 0  | 0.10 → 0.32     |

for the correct answer can no longer be assembled by any member following a controlled perturbation. Stable, adaptive and stressed states are defined by the same measure at differing levels of severity and recovery. Because the standard literature offers no such definition, producing a defensible one is treated as a contribution in its own right rather than a preliminary step.

3) *Separation of labels from features*: outcome labels derive exclusively from which required information items arrived and when; predictive features derive exclusively from the message log. This separation prevents the circularity that would otherwise inflate accuracy without establishing anything.

4) *An honest test of separability*: the study reports whether adaptation and breakdown can in fact be distinguished from early interaction structure. A negative result—that the two are not separable at the horizons examined—was planned for as a legitimate outcome, and it is the result obtained.

### III. METHODOLOGY

This study is conducted entirely in simulation. No human participants are involved, so no ethical approval is required. The trade-off is accepted deliberately: agent-based simulation permits the thousands of controlled runs that supervised learning requires, at the cost of psychological realism. Section V-C states the resulting limits.

The methodology below is the one implemented. Where it departs from the original proposal, the departure and its reason are stated at the point where it occurs. Parameter values are collected in Table III.

#### A. Research Design

The experiment crosses two factors, giving four primary conditions: group composition (human-only, human plus AI) by stress level (calm, stressed). Stress is applied identically across composition conditions, so AI presence is the only variable distinguishing them.

A fifth control condition contains one additional human agent in place of the AI agent. Without it, any difference between composition conditions could be attributed to group size rather than to the presence of an artificial participant. C4 and C5 therefore have the same number of agents and the same number of items, and differ only in what the thirteenth participant is. Table II summarises the design. The design has no size control under calm conditions; a supplementary check that supplies one is reported in Section IV-B.

#### B. Task Environment

Each run instantiates a hidden-profile task [10]. Every agent, the AI included, starts with three information items, so a group

of twelve humans holds 36 and a group of thirteen agents holds 39. Twelve of those items are drawn at random as the required set. The correct option becomes identifiable when all twelve reach a single agent. No agent starts with more than three, so no agent can reach the answer alone. Items are integer identifiers; the simulation carries no natural-language content, and the semantic framing exists only in the write-up.

The proposal specified that each item be held by more than one agent from the start, so that removing an agent could never destroy information. The implementation does the opposite: every item has exactly one holder at initialisation, and copies exist only once agents have passed them on. The reason is that redundancy handed out in advance would make a group's resilience to losing a member a property of the set-up rather than of the group. With a single initial holder, how much it has already shared, which is the group's own doing. The cost of this choice is that a perturbation can destroy a required item outright, and Section IV shows how much that cost matters.

### C. Agent Model

A human agent is defined by three state variables: the set of items it holds, a vector of trust values toward every other agent, and a scalar load in  $[0, 1]$  representing cognitive burden.

Each run lasts 1,000 ticks. At every tick the living agents act once each, in random order. An agent  $s$  with load  $L_s$  transmits with probability

$$p_{\text{send}} = r(1 - L_s), \quad r = 0.30, \quad (1)$$

choosing one of its items uniformly at random. The recipient is drawn from all other agents with probability proportional to  $s$ 's trust in each,  $T_{sj}$ . The recipient  $j$  accepts with probability

$$p_{\text{accept}} = T_{js}(1 - L_j)w_s, \quad (2)$$

where  $w_s$  is the AI trust weight if  $s$  is the AI and 1 otherwise. An accepted item is added to the recipient's holdings; agents have no capacity limit. After each attempt, the sender adjusts its trust in that recipient: up by 0.05 after an acceptance, capped at 1, and down by 0.02 after a refusal, floored at 0.05. Trust starts at 0.50 in every direction. An agent's trust in a partner therefore changes only through its own attempts to send to that partner.

The asymmetry between the increment and the decrement was introduced during calibration. With symmetric steps, trust rises only if more than half of attempts succeed. Under load that rate was out of reach, trust drained to the floor in every condition, and no group functioned. The floor is above zero so that a route that has gone quiet can still be tried again.

Recovery after perturbation is not implemented as a rule. No instruction directs an agent to compensate for a lost member. Recovery, where it occurs, emerges from trust updating and trust-weighted partner selection: failed transmissions to an absent agent lower its trust score, and remaining agents consequently redistribute traffic toward reachable partners. Whether this suffices to restore information flow depends on prevailing load, which is precisely the effect under study.

The AI agent differs in exactly three respects, and these constitute the operational definition of an artificial participant in this model:

1) *Load immunity*: its load stays at the calm value of 0.10 whatever the stress schedule, reflecting that computational systems do not experience social or cognitive stress [12].

2) *Imperfect reliability*: on 15 per cent of its transmissions the AI sends a well-formed item identifier that it does not hold, drawn from a reserved range that can never collide with a genuine item. Recipients cannot tell the difference, and such items never count towards integration. This operationalises the mechanism suggested by the meta-analytic finding that human-AI combinations frequently underperform the stronger party alone [2].

3) *Distinct trust weighting*: acceptance of AI-originated items is multiplied by a separate weight  $w$ , permitting over- and under-reliance to be modelled independently of inter-human trust [13]. In this study  $w = 1$ , the neutral value; sweeping it is left to future work.

Because agents have no capacity limit and items carry no content, a false item costs the group only the transmission it occupied. It cannot displace a genuine item or mislead a recipient. The AI's unreliability is therefore a weak manipulation in this implementation, and Section V returns to what that means for the AI results.

### D. Stress Schedule

Stress is operationalised as the load parameter of human agents. Calm conditions hold load at 0.10 for the full run. Stressed conditions start at the same 0.10, rise linearly between ticks 10 and 50 to 0.32, and hold that value to the end. The schedule is identical in C2, C4 and C5.

Because load reduces both sending and accepting, its effect on a single exchange compounds: at peak load a human-to-human exchange succeeds at  $(0.68/0.90)^2 \approx 0.57$  of its calm rate, before any change in trust. The proposal did not fix a peak value. An earlier setting of 0.45 was rejected during calibration because it stalled stressed groups short of the required set so reliably that the adaptive state was left almost empty.

This is a deliberate simplification and is claimed as such. Load is not a model of human stress physiology; it is a single controlled parameter that degrades transmission and reception, following the recommendation that stress manipulations be separated from time pressure rather than conflated with it [8].

### E. Perturbation

At tick 55 in every run, five ticks after the stress ramp completes, one human agent is chosen uniformly at random and removed. Items held only by that agent become unreachable; items it had already passed on remain in circulation. Remaining agents receive no notification and continue operating under unchanged rules. They keep choosing the absent agent as a recipient, every such attempt fails, and trust in that route falls until traffic moves elsewhere.

The perturbation is held constant across all conditions so that differences in recovery are attributable to condition

rather than to disturbance type. Its purpose follows from early-warning-signal theory: systems approaching a critical transition recover more slowly from disturbance than systems that are merely perturbed [21]. Since adaptation and breakdown resemble one another at the moment of disruption, the discriminating information lies in what follows.

In human+AI conditions the removed agent is always human, so that AI presence is held constant across the perturbation.

One consequence of the random choice needs stating plainly, because it turns out to dominate the results. If the removed agent was, at tick 55, the only holder of any required item, the correct option can never be assembled in that run.

## F. Outcome Labelling

Labels are derived exclusively from the information state and never from the communication record. This separation is the central methodological commitment of the study: since the predictive features are drawn from the communication record, defining outcomes in those same terms would make the prediction task circular and inflate accuracy without establishing anything.

*Integration*  $I(t)$  is the number of required items held by the best-informed living agent at tick  $t$ , out of  $N = 12$ . The AI's holdings count, since it is a member of the group; false items never count. Let  $t^*$  be the first tick at which  $I(t) = N$ , if there is one. Four states are assigned, checked in this order:

- **Broken down:** a required item was destroyed at the perturbation, or the run ends with  $I(T) < 0.6N$ . The correct option is never identifiable.
- **Stressed:** nothing was destroyed and  $I(T) \geq 0.6N$ , but  $I(t)$  never reaches  $N$ . Partial arrival continues without completion.
- **Stable:**  $t^* \leq 450$ . The group completes the task early, and the perturbation does not visibly delay it.
- **Adaptive:**  $t^* > 450$ . The group completes the task, but only late, after absorbing the loss.

This is a departure from the proposal, which defined recovery as the rate of required-item arrival returning to a fraction of its pre-perturbation level. Calibration showed that measure to be unusable. Arrival slows as the best desk fills, whatever else happens, because fewer required items remain to arrive. The smoothed arrival rate peaked during the first 30 ticks and then fell steadily in calm and stressed runs alike, so it never “returned” in either and could not separate them. Time to completion captures the same idea—how long a group takes to recover full integration after losing a member—without depending on the shape of a saturating curve.

Following the requirement that simulated datasets carry no evidential weight absent a pre-specified validation strategy [23], both thresholds (tick 450 and the 60 per cent floor) were fixed on calibration seeds and were not revised after any model was trained. Completion times form a single broad hump, rising from tick 182 to a peak near tick 575 and then tailing off flat to the end of the run, with no gap anywhere

that would justify placing the stable threshold in it. Sensitivity across a grid of both thresholds is reported instead.

## G. Calibration

Seeds 0 to 299 were reserved for calibration and never enter the study dataset. Five checks were run on them before any study data were kept: whether stress produced a visible gap in integration between C1 and C2; whether all four states were populated; whether the AI was so often the best-informed agent that “the group integrated the information” would really mean “the AI did”; whether the calm AI effect survived an added calm size control; and how much redundancy groups had built by the perturbation tick.

Three provisional values were changed as a result. The required set was reduced from 16 items to 12, the run was lengthened from 800 ticks to 1,000, and the peak load was lowered from 0.45 to 0.32. At the provisional values stressed groups almost never completed, which left the adaptive and stable states nearly empty under stress. The revised values were chosen by sweeping the required set (12–16), the run length (800–1,400) and the peak load (0.26–0.45) on calibration seeds and taking the combination under which every state appeared somewhere in the design while the stress effect remained clear. A first study dataset had been generated at the provisional values; it was discarded when the values changed. No classifier had been trained on it.

## H. Feature Extraction

Features are computed exclusively from the message log over an initial window of each run, so that prediction precedes the outcome being predicted. The primary window is ticks 0–199, the first fifth of the run. It contains the perturbation at tick 55 and the first 145 ticks of the group's response. A window ending at the perturbation would contradict the argument of Section II-F, that adaptation and breakdown differ in what follows the disruption rather than in the disruption itself. A window much longer would start to overlap the outcome: in the study data only 5 of 5,000 runs completed the task inside the first 200 ticks. The window length was chosen on calibration seeds, and the prediction-horizon analysis in Section III-J reports every shorter window as well.

Five families are extracted:

- 1) *Volume*: total transmissions, mean transmissions per tick, and the slope of the transmission rate.
- 2) *Distribution*: Gini coefficient of transmissions per agent, count of silent agents, and the ratio of most to least active agent.
- 3) *Structure*: number of connected components in the interaction graph, graph density, and degree centralisation [15], [16].
- 4) *Reciprocity*: proportion of transmissions followed by a transmission in the reverse direction, and mean delay to reciprocation.
- 5) *Instability*: rolling variance of transmission rate and lag-1 autocorrelation of transmission rate, these being the observable signatures of critical slowing down [21].

In human+AI conditions, two further features record the proportion of AI-originated items accepted and the ratio of AI-to-human against human-to-human traffic. Both are zero in human-only runs.

No feature references item identity, integration, or task outcome. This is enforced rather than merely checked: before extraction, each logged message is reduced to five fields—tick, sender, recipient, whether it was accepted, and whether the sender was the AI—so the extraction code never receives the item identifier or the flag marking a false item.

Three of the fifteen features were dropped before modelling. The count of silent agents was 0 and the number of connected components was 1 in every run: at these parameters every agent speaks and the interaction graph is always connected within the window. Mean transmissions per tick is the total divided by a fixed window length, so it duplicates the total exactly. Twelve features remain.

## I. Predictive Modelling

Each run yields one fixed-length feature vector and one label, so the prediction task is multi-class classification over tabular data rather than sequence or graph learning. Although temporal graph architectures exist for evolving networks [20], they address dynamic link prediction rather than whole-graph state classification, and applying them to a fixed-length summary per run would add complexity without a corresponding gain in what can be concluded.

Two classifiers are therefore trained: multinomial logistic regression on standardised features, selected for interpretability of coefficients, and a random forest of 300 trees with at least five runs per leaf, selected for its tolerance of non-linear feature interactions. Both weight classes inversely to their frequency, since the stable state is rare. Feature importance is measured by permutation on the held-out set, as the drop in macro-F1 when one feature is shuffled; this serves objective three directly.

Two baselines were planned. A majority-class predictor establishes the floor. A single-feature logistic regression on transmission volume alone establishes whether the structural and instability features contribute anything beyond a naive measure of activity. If the full models do not exceed this second baseline, that result is reported as the finding.

Two comparisons were added once the design was implemented. A *condition-only* model is a logistic regression on the five condition indicators and nothing else. Stress changes message volume substantially, so a feature model can score well merely by recognising whether a run was stressed; the condition-only model measures how far that alone goes. An *information-state ceiling* fits the same random forest to four quantities the predictors are forbidden to see, all known by the end of the feature window: integration at tick 199, the number of items still held by a single agent at the perturbation, how many of the removed agent’s original items had not been shared, and whether a required item was destroyed. No model of the message log can be expected to beat this ceiling, so it

indicates how much of the outcome is settled by tick 200 at all.

## J. Evaluation

The 5,000 runs are split 80/20 into training and held-out test sets, stratified by label, with a fixed split seed. Five-fold stratified cross-validation is run on the training partition, and the held-out set is touched once per model. Reported metrics are accuracy, macro-averaged F1, per-class precision and recall, and the full confusion matrix. Macro-averaging is specified because class frequencies differ across conditions and overall accuracy would otherwise be dominated by the majority state.

Four analyses go beyond headline accuracy. The first three were planned; the fourth was added for the same reason as the condition-only baseline.

1) *Adaptive versus broken-down separability.* These two states are the difficult pair, since both involve structural reorganisation following disturbance. Binary classification restricted to these classes is reported separately, as aggregate accuracy would otherwise conceal failure on the case of interest.

2) *Prediction horizon.* Models are retrained on features computed over windows ending at ticks 55, 100, 150 and 200, to establish how early a state becomes predictable and whether seeing the response to the perturbation adds anything. The window ending at tick 55 contains no post-perturbation data at all.

3) *Threshold sensitivity.* Labels are recomputed for every combination of stable threshold (ticks 350, 450, 550) and breakdown floor (60, 70, 80 per cent) and the models retrained. A conclusion that holds only at one threshold value is reported as an artefact rather than a finding.

4) *Within-condition prediction.* Models are cross-validated inside each condition separately, where recognising the condition cannot help, and compared with a classifier that guesses in proportion to class frequency.

## K. Implementation

The simulation is implemented in Python with deterministic seeding, so that any run can be reproduced exactly and any observed transition traced to the rule that produced it. Classifiers use scikit-learn. One script generates the dataset and writes the predictors, the labels and the information-state quantities to separate files; a second trains and evaluates every model reported below. Re-running both regenerates every number in Section IV.

# IV. RESULTS

## A. The Dataset

Table IV gives the share of runs in each state. Stress had the effect the study depends on. Under calm conditions a quarter of human-only groups completed early and almost none stalled; under stress early completion all but disappeared, falling to about one per cent, and a quarter to a third of runs ended

TABLE III  
SIMULATION AND EVALUATION PARAMETERS

| Parameter                   | Value              | Range tested                |
|-----------------------------|--------------------|-----------------------------|
| Human agents per group, $n$ | 12                 | 12–13                       |
| Items per agent (total)     | 3 (36 or 39)       | —                           |
| Required items, $N$         | 12                 | 12–16 <sup>a</sup>          |
| Holders per item at start   | 1                  | —                           |
| Ticks per run               | 1,000              | 800–1,400 <sup>a</sup>      |
| Runs per condition          | 1,000              | —                           |
| Base transmission rate $r$  | 0.30               | —                           |
| Load, calm                  | 0.10               | —                           |
| Load, stressed start / peak | 0.10 / 0.32        | peak 0.26–0.45 <sup>a</sup> |
| Stress ramp                 | ticks 10–50        | —                           |
| AI false-item rate          | 0.15               | 0–0.15 <sup>b</sup>         |
| AI trust weight $w$         | 1.0                | —                           |
| Initial trust               | 0.50               | —                           |
| Trust up / down / floor     | 0.05 / 0.02 / 0.05 | —                           |
| Perturbation tick           | 55                 | —                           |
| Feature window              | ticks 0–199        | 55–200                      |
| Stable threshold on $t^*$   | 450                | 350–550                     |
| Breakdown floor, $I(T)/N$   | 0.60               | 0.60–0.80                   |
| Train / test split          | 80 / 20            | —                           |
| Cross-validation folds      | 5                  | —                           |
| Calibration seeds           | 0–299              | —                           |
| Study seeds                 | 10,000–14,999      | —                           |

<sup>a</sup>On calibration seeds only. <sup>b</sup>Supplementary check, Section IV-B.

TABLE IV  
SHARE OF RUNS IN EACH STATE (1,000 RUNS PER CONDITION)

| Condition                 | Stable | Adaptive | Stressed | Broken |
|---------------------------|--------|----------|----------|--------|
| C1 Human-only, calm       | 0.259  | 0.488    | 0.000    | 0.253  |
| C2 Human-only, stressed   | 0.011  | 0.425    | 0.257    | 0.307  |
| C3 Human+AI, calm         | 0.165  | 0.599    | 0.010    | 0.226  |
| C4 Human+AI, stressed     | 0.008  | 0.382    | 0.355    | 0.255  |
| C5 Size control, stressed | 0.008  | 0.402    | 0.319    | 0.271  |
| All runs                  | 0.090  | 0.459    | 0.188    | 0.262  |

stalled short of the full set. Stress also raised the rate of breakdown in human-only groups from 25.3 to 30.7 per cent.

Every one of the 1,312 broken-down runs was broken down for the same reason: the removed agent had been the only holder of at least one required item. The 60 per cent floor on final integration never decided a label. This is the single most important fact about the dataset. It means the broken-down state records a loss at tick 55, not a failure to integrate afterwards. Stress raises the rate of such losses indirectly. By the perturbation tick, calm human-only groups had left 9.9 items on average with a single holder, and stressed ones 11.8, because the stress ramp had already cut how much they passed on.

## B. Effect of the AI Participant

Under stress the AI made little difference. C4 and C5 differ by at most 3.6 percentage points in any state, and each share has a standard error of about 1.5 points at 1,000 runs, so none of those differences is clearly outside chance. The largest is a higher share of stalled runs with the AI (35.5 against 31.9 per cent).

Under calm conditions C3 completed early less often than C1 (16.5 against 25.9 per cent). Because the design has no calm size control, a supplementary check ran 1,000 further runs of each of four calm configurations on seeds 20,000–

TABLE V  
FOUR-CLASS RESULTS. CV: MEAN  $\pm$  SD MACRO-F1 OVER FIVE FOLDS OF THE TRAINING SET. TEST: HELD-OUT 20 PER CENT

| Model                     | CV F1             | Test F1 | Test acc. |
|---------------------------|-------------------|---------|-----------|
| Majority class            | 0.157 $\pm$ 0.000 | 0.157   | 0.459     |
| Volume only (logistic)    | 0.274 $\pm$ 0.007 | 0.249   | 0.257     |
| Condition only (logistic) | 0.205 $\pm$ 0.003 | 0.205   | 0.271     |
| Logistic regression       | 0.302 $\pm$ 0.007 | 0.316   | 0.323     |
| Random forest             | 0.319 $\pm$ 0.011 | 0.341   | 0.353     |
| Random forest + condition | 0.326 $\pm$ 0.013 | 0.348   | 0.359     |
| Info-state ceiling        | 0.599 $\pm$ 0.011 | 0.641   | 0.654     |
| without destroyed flag    | 0.416 $\pm$ 0.009 | 0.457   | 0.469     |

20,999, outside both the calibration and study ranges. Early completion was 26.6 per cent with twelve humans, 19.2 per cent with thirteen, 18.6 per cent with twelve humans and an error-free AI, and 16.1 per cent with the standard AI. Most of the drop is therefore an effect of group size. A thirteenth participant, human or not, adds three items to the pool and one more partner for every sender to divide its attention between, which probably explains the slower completion. The AI's errors account for a further two to three points, which is at the edge of what 1,000 runs can resolve.

One difference does favour the AI conditions, and it is mechanical. The AI is never removed, so required items it holds can never be destroyed. Human+AI groups broke down less often than human-only groups at the same stress level (22.6 against 25.3 per cent calm, 25.5 against 30.7 per cent stressed).

## C. Four-Class Prediction

Table V reports the main comparison. Both feature models beat the majority-class floor and the volume-only baseline, which was the planned test of whether structure adds anything to activity. The random forest reached a held-out macro-F1 of 0.341 against 0.249 for volume alone.

The per-class results show where that score comes from. The forest recovered 61 per cent of stable runs and 83 per cent of stressed runs, but only 20 per cent of adaptive runs and 20 per cent of broken-down ones. Of the 459 adaptive runs in the test set it labelled 187 as stressed and 122 as stable; of the 263 broken-down runs it labelled 121 as stressed. Stable runs occur almost only under calm conditions and stressed runs almost only under stress, so these are the two states that knowing the level of stress gives away. The logistic regression showed the same pattern more sharply, with recall of 80 and 78 per cent on stable and stressed and 11 and 21 per cent on adaptive and broken down.

Adding the condition to the forest's inputs raised its test score by only 0.007, which suggests that the features already encode most of what the condition carries.

## D. Within One Condition

When models were cross-validated inside a single condition, where recognising the condition is no help, they came close to chance. Against a classifier that guesses in proportion to class frequency, the forest's macro-F1 was between 0.006 below and

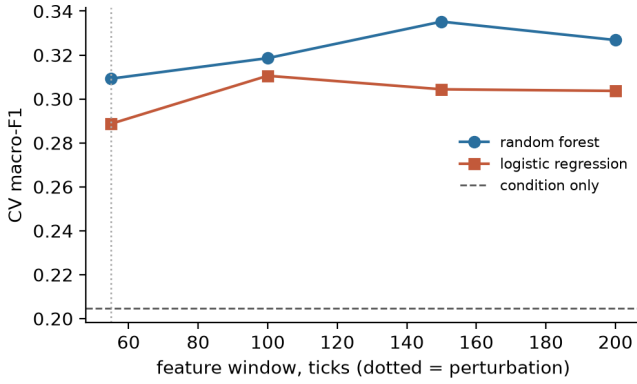


Fig. 1. Cross-validated macro-F1 on the training set by length of the feature window. The dotted line marks the perturbation at tick 55; the dashed line is the condition-only baseline.

0.045 above chance across the five conditions, and the logistic regression’s between 0.030 below and 0.050 above. The largest gains, both in C1, were 0.045 and 0.050. Whatever the features know about a run’s fate beyond its condition, it is little.

### E. Adaptive Versus Broken Down

Restricted to the two states the study was most interested in separating, the held-out macro-F1 was 0.578 for logistic regression and 0.560 for the forest, against 0.525 for condition alone and 0.389 for the majority class. The information-state ceiling separated the pair perfectly, which is simply the destroyed-item flag restating the definition of breakdown. Once that flag is known, adaptation and breakdown are trivially distinct. Without it, the message log contains only a faint trace of the difference.

### F. Prediction Horizon

Fig. 1 shows cross-validated macro-F1 as the feature window lengthens. With a window that ends at the perturbation the forest scored 0.309; with 95 ticks of the response included it reached 0.335. Every window scored above the condition-only line. Seeing how a group responds to the loss therefore helps, but only by about 0.03, and the gain levels off after tick 150.

### G. Threshold Sensitivity

Across all nine combinations of stable threshold and breakdown floor the forest scored between 0.319 and 0.362, and stayed between 0.09 and 0.19 above the condition-only model. Changing the floor from 60 to 80 per cent moved at most 0.3 per cent of runs, because breakdown is set almost entirely by destruction. Moving the stable threshold shifted runs between the stable and adaptive states: the stable share was 2.8 per cent at tick 350 and 18.7 per cent at tick 550. The conclusions above hold at every setting tested.

### H. Feature Importance

Permutation importance on the held-out set ranked total transmissions first (a drop in macro-F1 of  $0.020 \pm 0.012$ ), followed by the ratio of most to least active agent ( $0.016 \pm 0.008$ )

and the two AI features ( $0.013 \pm 0.005$  and  $0.008 \pm 0.005$ ). The AI features are zero in every human-only run, so their importance is largely the AI conditions identifying themselves. No structural or instability feature—density, degree centralisation, rolling variance, lag-1 autocorrelation—had an importance distinguishable from zero. Shuffling reciprocity improved the score slightly ( $-0.016 \pm 0.008$ ), which indicates the forest had fitted noise in it.

## V. DISCUSSION

### A. Why Adaptation and Breakdown Did Not Separate

The central question was whether early interaction structure indicates whether a group will adapt or break down. In this model it largely does not, and the reason can be traced to a single rule. Breakdown occurred when the agent removed at tick 55 was the only holder of a required item. The removed agent is chosen at random. A group’s communication before tick 55 decides how many items are still exposed—stressed groups left more—but not which agent is lost. Even a perfect observer of the message log could learn only the odds of breakdown, not whether it would happen.

This was checked directly rather than assumed. On 300 further runs per condition, the state of every group at tick 55 was used to compute the probability that a random removal would destroy a required item. That probability averaged 0.27 and the realised breakdown rate was 0.28, so the rule is behaving as described. It was almost never close to 0 or 1 (in 1 per cent of runs), which means nearly every group was exposed and the draw alone decided which ones paid. Used as a score, it separated broken-down runs from the rest with an AUC of only 0.65. Its squared error was 0.187, against 0.186 expected from the draw alone and 0.199 for always guessing the base rate. Even an observer who sees every desk in the group, and not merely the message log, therefore has almost no information beyond the odds. The result does not say that the simulation made no progress: stress effects on completion are large and consistent. It says that the breakdown label, as designed, is mostly a coin flip.

The information-state ceiling makes the same point from the other side: it reached 0.641 only because it was handed the destroyed-item flag, and fell to 0.457 without it.

This is a negative result, and the design planned for it. What it shows is narrower than “adaptation and breakdown cannot be told apart.” It shows that when disruption strikes a random member, and the group has built little redundancy by then, breakdown is mostly luck, and a prediction problem built on it has a low ceiling for any method. That is itself a claim about groups, if a modest one: a team whose knowledge is concentrated can be undone by who it happens to lose, and no amount of observation of its communication will warn of that in advance.

### B. Why the Early-Warning Features Were Silent

The instability features were chosen because critical slowing down precedes transitions in many systems [21]. They carried no signal here, and that is to be expected in hindsight. Critical

slowing down is a property of a system approaching a tipping point. Integration in this model is an accumulation: items are added and never lost except at the perturbation, and there is no state for the group to tip out of. The model’s dynamics contain nothing for rising variance or autocorrelation to anticipate. The qualification of Rietkerk et al. [22], that such signals do not distinguish harmful from harmless transitions, never came into play, because the model contains no transition of either kind for them to precede.

### C. Threats to Validity

*Agent realism.* Rule-based agents have no emotion, misunderstanding, or status concern. The study models the structure of information flow under load, not human psychology, and results transfer to human groups only as hypotheses requiring separate empirical test. The model is a prototype of that test. Its simulated humans are also the ones being removed, so the outcome rests on a random draw over agents whose only differences are the items they happen to hold; real members differ in role, memory and habit, and whether they differ in ways that make loss predictable is an empirical question this model cannot answer.

*Self-referential validation.* Both data and labels originate in the same simulation. The label–feature separation, the reserved calibration seeds and the sensitivity analysis constrain but do not eliminate this circularity [23]. No claim of external validity is made.

*Parameter dependence.* The proposal intended to establish findings across a range of group sizes and item distributions. That was done only during calibration and in the supplementary size check; the study dataset uses one configuration. Results that depend on it, such as the exact share of broken-down runs, should not be generalised.

*A single kind of disruption.* Only the random removal of one member was tested. Section V-A suggests that this choice, more than any other, determined the outcome.

*A weak AI manipulation.* Because false items occupy only a transmission and cannot crowd out or contradict genuine ones, the AI’s unreliability had little room to matter. The absence of a clear AI effect under stress should not be read as evidence that unreliable AI is harmless to stressed groups.

## VI. CONCLUSION AND FUTURE WORK

This study built an agent-based model of a group pooling information under stress, with and without an artificial participant, and asked whether the group’s early message log indicates whether it will adapt to losing a member or break down. Stress had a clear and consistent effect on outcomes. Classifiers on the message log beat all planned baselines, but most of what they learned was the level of stress, and within a condition they were close to chance. Every breakdown traced to the random loss of a member who held something no one else had.

Four changes follow directly. *Targeted perturbation:* removing the most active or best-informed member, rather than a random one, would make breakdown depend on structure that the

features can observe, and would test whether concentration and centralisation become predictive. *Recovery beyond completion:* labels based on time to completion conflate slow groups with disrupted ones; comparing each run with a run from the same seed in which no one is removed would isolate the effect of the loss. *A stronger AI manipulation:* capacity-limited agents, or false items that contradict genuine ones, would give the AI’s errors a cost, and the trust weight could then be swept to model over- and under-reliance. *Parameter sweeps:* group size, items per agent and the timing of the perturbation should be varied in the study data itself, not only during calibration.

### A. The Case for Volunteer Studies

The prototype is useful mainly for what it lets a human study skip. Labels, thresholds, the 5,000-run feature pipeline and a sensitivity analysis are written and frozen, so a volunteer study can be pre-registered with its analysis already built. Its effect sizes give a planning basis: stress cut early completion from about 26 to 1 per cent, so that effect is large enough to detect with a few dozen groups, whereas the 5-point difference in breakdown with and without stress would need several hundred. Four reasons make volunteers the right next step.

*The weakest point is the one volunteers change.* Breakdown here is a draw over rule-based agents with no capacity to anticipate loss. People rehearse, repeat, note who knows what and re-share before they are told to. Whether groups of volunteers build the redundancy this model fails to build, and whether that makes breakdown avoidable and therefore predictable, is the open question, and only people can answer it.

*Stress becomes a manipulation, not a parameter.* Time pressure, noise or competing tasks can be applied to real members, separated from one another as recommended in [8], and compared against a calm control, which a load parameter cannot do.

*The perturbation can be made causal and fair.* Random disconnection of one volunteer, or a targeted one chosen by activity or by what they hold, is an assignment the experimenter controls. Pairing each stressed group with an unperturbed twin, as suggested above, separates the effect of the loss from slow completion.

*The AI can be given a real cost.* A scripted assistant that is wrong a fixed share of the time, trusted or distrusted by the volunteers themselves, replaces the assumed trust weight with a measured one, and answers whether the penalty reported for human–AI teams [2] grows under stress.

Such a study does not need the model to be right. It needs the model to have asked a well-formed question, and to have shown which parts of the question a simulation can settle and which it cannot. Results from volunteers should be reported against this prototype’s negative result, whichever way they go.

The larger aim stands: to treat collective intelligence as a trajectory and to define failure carefully enough to predict it. What this study adds is a worked example of how the definition of failure can decide the answer before any model is trained.

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